

SUDARSHANA KARKALA

Director of Engineering | Technology & R&D Consultant

Space Systems & Applications · Avionics & Telemetry

EV Battery & Energy Intelligence · AI · Cybersecurity · Digital Twins

Bengaluru, India | sudarshana.karkala@gmail.com | +91 9845561518 | [LinkedIn](#) | Consulting & Advisory

Exploring Director of Engineering, Technology Consulting and R&D opportunities across Space, Aerospace and mission-critical EV/Energy systems.

SUMMARY

- Engineering and technology leader with **20+ years of experience** across software architecture, cybersecurity, connected systems, IoT, telematics, AI, mobility, EV and energy platforms.
- Led global engineering teams and production-scale platforms spanning device connectivity, telemetry, cloud, security and intelligent diagnostics.
- Currently expanding R&D focus into Space Systems & Applications, CanSat Model Rocketry, Avionics & Telemetry and Aerospace Cybersecurity, applying established systems-engineering experience to aerospace and space technology.

STRATEGIC TECHNOLOGY FOCUS

- **Space Systems & Applications:** Space Technology R&D · Space Applications · Mission/Ground Systems Concepts · Digital Twins · AI/Data Applications
- **Avionics & Telemetry:** CanSat/Model Rocket Avionics · Sensors · Telemetry · CAN · MQTT · IoT · Secure Device Communication · Ground Systems
- **EV Battery & Energy Intelligence:** BMS · Battery Safety · Diagnostics · Digital Twin · Predictive Maintenance · Battery Cybersecurity · EVSE
- **AI & Cybersecurity:** GenAI · Agentic AI · Predictive Analytics · RAG · Connected-System Security · Automotive/Aerospace Cybersecurity
- **Engineering Leadership:** Technology Strategy · Systems Architecture · Engineering Organisations · R&D Programs · Product Engineering · Technical Roadmaps

SELECTED ACHIEVEMENTS

- Led **30+ engineers** delivering global Solar, EVSE, Energy and IoT platforms at Enphase.
- Architected mobile/security platforms serving **3M+ users**, achieving a **4.7★ App Store rating** at McAfee.
- Architected AI-powered EV Battery Safety, Battery Digital Twin, Battery Aadhaar and EV Telematics platforms.
- Delivered secure connected systems spanning IoT, BLE, CAN, MQTT, cloud, telemetry and cybersecurity.
- Mentors engineering R&D initiatives across EV Battery, CanSat Model Rocketry, Aerospace Cybersecurity and Space Technology.

CURRENT ROLE

Thasmai Infotech Private Limited | Bengaluru | Nov 2025 – Present

Co-Founder & Director of Engineering — EV Battery Intelligence, AI & Engineering R&D

Leading technology architecture and R&D initiatives across EV Battery Intelligence, AI, connected engineering systems and cybersecurity.

- Architected an AI-driven EV Battery Safety Platform for thermal-risk prediction, diagnostics and predictive maintenance.
- Developed architectures for Battery Digital Twin, Battery Aadhaar, BMS intelligence and battery cybersecurity.
- Designed connected EV platforms integrating battery telemetry, BLE/MQTT, cloud, security and intelligent diagnostics.
- Leading and mentoring emerging R&D initiatives across EV, aerospace and space technologies.

Space & Aerospace R&D | EV.ENGINEER™ | EV Society™ | 2026 – Present

Space Systems & Applications · CanSat Model Rocketry · Avionics & Telemetry · Aerospace Cybersecurity

- Exploring Space Systems & Applications and CanSat Model Rocketry through engineering R&D, workshops and student research programs.
- Working with model-rocketry concepts covering system design, simulation, avionics, telemetry, recovery, safety, testing and engineering documentation.
- Exploring applications of AI, telemetry, digital twins, connected systems and cybersecurity for aerospace and space systems.
- Mentoring student teams through structured requirements → design → simulation → build → test → review engineering workflows.

Space Research: <https://aerospace.ev.engineer/aerospace> | <https://aerospace.ev.engineer/space>

Workshop Gallery: <https://autonomous.ev.engineer/workshop-gallery>

SPACE TECHNOLOGY PROFESSIONAL DEVELOPMENT

Essentials of Model Rocketry | IN-SPACe | Aug 2026

- Skill-development workshop under IN-SPACe, Department of Space, Government of India, conducted at GNEC, IIT Roorkee — Greater Noida Campus.
- **Focus:** CanSat Model Rocketry · Avionics & Telemetry · Simulation · Aerospace Cybersecurity · Space Applications

PROFESSIONAL EXPERIENCE

Principal Architect / Consultant — eMobility, AI & Connected Systems

Bengaluru | Dec 2024 – Nov 2025

- Architected connected EV platforms integrating battery systems, telematics, BLE, cloud, security and digital twins.
- Designed secure device communication, battery-swapping intelligence and telemetry architectures.
- Delivered consulting across automotive, embedded engineering, IoT, AI and predictive-maintenance platforms.

Selected Engagements: Schaeffler India · Crompton Greaves Consumer Electricals · iTelematics · Skill-Lync · iFab Innovations

Enphase Solar Energy Private Limited | Bengaluru | Jul 2021 – Dec 2024

Sr. Engineering Manager — Mobile Apps, EVSE, IoT & Energy

- Built and led engineering organisations of 30+ engineers across Mobile, IoT, EVSE and connected-energy systems.
- Led architecture and delivery of global device commissioning and energy platforms.
- Designed secure BLE, Wi-Fi, gateway and cloud-connected systems.
- Led engineering strategy, architecture governance, roadmaps, product delivery and cross-functional engineering execution.

McAfee Software (India) Private Limited | Bengaluru | Oct 2011 – Jul 2021

Engineering Manager — Mobile Security & Telematics

- Architected and developed McAfee Mobile Security (iOS), reaching 3M+ users and 4.7★ App Store rating.
- Developed Telematics Security applications using OBD-II and connected-vehicle data.
- Led architecture across security, encryption, privacy, telemetry and connected systems.
- Managed product engineering, technical roadmaps and engineering teams.

Earlier Engineering Experience | 2005 – 2011

eNovate Media Solution · Ness Technologies · NMS Communications

Technical leadership and software engineering across mobile platforms, airline/travel applications, communications, multimedia and connected systems.

EDUCATION & PROFESSIONAL DEVELOPMENT

- B.E. — Information Technology
National Institute of Technology Karnataka (NITK), Surathkal | 2001–2005
- Electric Vehicle Engineering & Development
IIT Madras - CODE | 2024–2026
- Essentials of Model Rocketry
IN-SPACE, Department of Space, Government of India | GNEC, IIT Roorkee, Greater Noida | Aug 2026
- Certified Scrum Product Owner (CSPO)
- Certified EV Technology Officer (EVTO™) — **EV Society™** | In Progress | Expected Mar 2028
- Business P.A.C.E. Program — Rajiv Talreja / QuantumLeap

CONSULTING & R&D COLLABORATION

Available for **Director of Engineering, Technology Consulting, Systems Architecture, R&D and strategic collaboration** in:
Space Systems & Applications · Avionics & Telemetry · EV Battery & Energy Intelligence
Aerospace Cybersecurity · CanSat Model Rocketry

- Professional Profile: <https://aerospace.ev.engineer/about/sudarshana-karkala>
- Resume: <https://carsoftwaresystems.com/public/sudarshanakarkala.pdf>
- LinkedIn: <https://www.linkedin.com/in/sudarshanakarkala>